

Low-Cost Servo-Actuated Autonomous Water Surface Cleaning Robot with Real-Time Turbidity Monitoring for Temple Pond Hygiene Maintenance

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Abstract — This paper presents the design, implementation, and experimental validation of WaterBot, a low-cost autonomous water surface cleaning robot for open water body hygiene maintenance. The system is realised on an Arduino Nano (ATmega328P, 16 MHz) microcontroller and integrates an analog infrared turbidity sensor for real-time water clarity assessment and dual servo-actuated paddle arms for active mechanical debris collection into a front-mounted mesh bin. System behaviour is governed by a three-state finite state machine (FSM): IDLE (continuous turbidity monitoring), COLLECTING (threshold-triggered autonomous cleaning, NTU > 100), and MANUAL (serial command-driven operation). Experimental results on a controlled 60-litre test basin demonstrate 15.0–26.0% turbidity reduction (NTU) per five-minute cleaning cycle, analogue-to-NTU computation latency of approximately 120 μ s, motor actuation response within 15 ms, and servo response within 350 ms. The complete prototype is realised at a hardware cost below ₹5,000 (approximately US\$60), making it the first known system to combine turbidity-triggered servo-actuation with autonomous aquatic debris collection at this price point. Systematic comparative analysis against six surveyed academic prototypes, five patents, and four commercial products confirms the system's originality and practical contribution.

Keywords — *autonomous robot; turbidity sensor; servo actuator; water surface cleaning; temple pond; Arduino Nano; ATmega328P; embedded systems; finite state machine; NTU*

I. INTRODUCTION

Open water bodies in semi-urban and rural India face persistent surface-level contamination from floating plastic waste, organic debris accumulation, and algae proliferation. Temple ponds (kulams or pushkarinis) across Tamil Nadu — numbering over 30,000 — are particularly vulnerable due to their open exposure, frequent human interaction, and continuous organic loading from ritual activities [1]. Over time, accumulated surface debris drives turbidity increase, oxygen depletion, and microbial growth, resulting in degraded water hygiene that affects groundwater recharge, irrigation, and occasional human contact.

Conventional maintenance of these water bodies relies on periodic manual cleaning, which is both resource-intensive and inconsistent. In the absence of regular intervention, contamination escalates from the surface layer — where organic decomposition and oxygen exchange occur — into the broader water column, making subsequent remediation progressively more difficult.

Commercial autonomous aquatic cleaning solutions, such as the WasteShark platform (RanMarine Technology, Netherlands), are priced at €10,000–€20,000, placing them entirely out of reach for temple trusts, local municipalities, or NGOs managing small water bodies. Academic prototypes, while demonstrating relevant technical capabilities, typically address either autonomous navigation, debris collection, or water quality sensing in isolation — not as an integrated system with sensor-driven mechanical actuation.

This paper presents WaterBot: a low-cost autonomous floating embedded platform built on the Arduino Nano microcontroller, integrating an analog infrared turbidity sensor for real-time water clarity assessment and dual servo-actuated paddle arms for active surface debris collection. The system targets the specific use case of small open water body maintenance — particularly temple ponds — where cost, simplicity, and field deployability are the dominant design constraints.

The primary contributions of this work are as follows:

- Integration of threshold-triggered turbidity sensing with servo-guided debris collection in a single embedded system below ₹5,000.
- A three-state FSM enabling autonomous, manual-override, and idle operating modes with safety timeout.
- Experimental validation demonstrating 15–26% turbidity reduction per cleaning cycle and sub-millisecond control latency on an ATmega328P platform.

The remainder of this paper is organised as follows: Section II reviews related work. Section III presents the system architecture and hardware design. Section IV details the embedded software and FSM. Section V presents calibration methodology and sensor validation. Section VI reports experimental results. Section VII provides comparative analysis. Section VIII concludes with directions for future work.

II. RELATED WORK

A. Academic Prototypes

The SMURF (Surface Marine Unmanned Robot for Floating waste) platform [2] demonstrates fully autonomous navigation and coverage path planning for floating debris collection using boustrophedon sweep patterns and obstacle avoidance. While effective for open-water navigation, the system does not integrate water quality sensing or threshold-based actuation — debris collection is continuous regardless of water state.

Jeon et al. [3] present an autonomous unmanned surface vehicle integrating water quality sensors (pH, dissolved oxygen, temperature) with GPS-based waypoint navigation for environmental monitoring. The system focuses exclusively on sensing without corrective mechanical action — no debris collection mechanism is included. This represents the complementary limitation to SMURF: sensing without collection versus collection without sensing.

The OSCAR (Ocean Surface Cleaning Autonomous Robot) system [5], developed at VIT Chennai, integrates embedded navigation with object classification and a front-mounted net for debris capture. A path planning algorithm optimises sweep coverage. However, the system does not incorporate water quality feedback, and corrective action is not triggered by sensor thresholds.

The autonomous lake cleaning robot documented in [6] employs a conveyor-belt collection mechanism with semi-autonomous control on an Arduino platform. While the cost profile is similar to this work, the system lacks sensor integration — turbidity and water quality are not measured — and manual intervention is required for navigation.

B. Commercial Products

WasteShark (RanMarine Technology) [4] is the closest commercial equivalent to this work, providing reliable autonomous debris collection at harbour and lake scale. Its payload capacity (16–40 kg of debris per cycle) and operational reliability are excellent. However, it carries no onboard environmental sensors, costs approximately €20,000 per unit, and is not designed for the shallow, small-area conditions of temple ponds.

Other relevant commercial systems include solar-powered skimmer products targeting swimming pools (Solar Skim Pro, ~₹1,50,000) and DJI aquatic drones (~₹80,000+) designed for inspection rather than cleaning. None of these systems integrate turbidity-triggered autonomous actuation at a price accessible for community-scale water body management.

C. Patents

US Patent 10,717,486 B2 [7] (Searobotics) claims an autonomous unmanned surface vehicle with sonar-based navigation and debris collection, targeting harbour environments. IN202041019765A [8] (IIT Bombay) addresses semi-autonomous aquatic weed removal with a

conveyor mechanism — specific to water hyacinth, not general floating debris. WO2019/053679A1 [9] (ETH Zürich) covers a multi-sensor aquatic robot with sensor fusion for pH, turbidity, and temperature combined with autonomous path planning — the most aligned existing patent, though focused on monitoring rather than corrective actuation and filed at research-prototype rather than product scale.

D. Research Gap

Across all surveyed work, no system simultaneously satisfies the following constraints: (1) real-time turbidity sensing with threshold-based mechanical actuation, (2) servo-actuated active debris guidance, (3) complete hardware cost below ₹5,000, and (4) targeting of small open water bodies such as temple ponds. This work directly addresses this gap.

III. SYSTEM ARCHITECTURE

A. Design Objectives and Constraints

The system design is governed by four primary constraints: (1) total hardware cost below ₹5,000; (2) operation on a compact floating chassis suitable for water bodies of 100–5,000 m²; (3) real-time turbidity monitoring with autonomous actuation response within one sensor sampling period (500 ms); and (4) safe field operation including manual override and fault-tolerant timeout logic.

B. Microcontroller — Arduino Nano (ATmega328P)

The Arduino Nano is selected as the central microcontroller following the failure of the originally planned ESP32-CAM module (camera damage during field testing). The ATmega328P operates at 16 MHz with 32 KB Flash program memory, 2 KB SRAM, 1 KB EEPROM, an 8-channel 10-bit ADC, six PWM output channels, and UART/SPI/I2C interfaces.

This system is classified as a Special Purpose Embedded System (SPES): the microcontroller, firmware, and hardware are co-designed for a single application. The Nano's power consumption (~19 mA active) is significantly lower than the ESP32-CAM (~240 mA with WiFi enabled) and the Raspberry Pi Zero (~230 mA idle), both of which were considered and rejected on power-budget grounds. Boot time on the Nano is sub-millisecond — critical for reliable initialisation on a floating platform where power-on conditions may be unstable.

Resource utilisation is efficient: the compiled firmware occupies approximately 5,800 bytes Flash (18% of available 32 KB) and 350 bytes SRAM (17% of available 2 KB), leaving substantial headroom for future sensor expansion.

C. Seven Functional Hardware Blocks

The system is organised into seven functional blocks as required by the project specification:

Block 1 — Power Supply: A 9V battery feeds a 7805 linear voltage regulator providing 5V regulated supply to the Arduino Nano, servos, and sensor module. Decoupling capacitors (100 μ F electrolytic + 100 nF ceramic) across the supply rails suppress switching transients from motor commutation. Total idle current draw is approximately 25 mA; peak current during motor + servo operation reaches approximately 1,000 mA.

Block 2 — Central MCU: Arduino Nano (ATmega328P) executing the FSM, ADC reads, servo PWM generation, motor direction/speed control, and UART serial data output.

Block 3 — Water Quality Sensing: Analog IR turbidity sensor connected to ADC pin A0. The sensor shines infrared light through a water sample; suspended particles scatter the beam, reducing the photodetector signal. The analog output (0–5V) is sampled at 10-bit resolution (0–1023 counts) and converted to NTU via a polynomial calibration curve.

Block 4 — Propulsion Subsystem: Two DC motors with propellers driven by an L298N dual H-bridge motor driver. Differential thrust (unequal speeds on left/right motors) achieves turning. Motor direction is controlled via digital GPIO; speed is regulated via PWM on the L298N Enable pins.

Block 5 — Servo Actuation: Two SG90 servo motors mounted on the port and starboard sides of the chassis drive paddle arms. The servos sweep $0^\circ \rightarrow 90^\circ \rightarrow 0^\circ$ in mirrored opposition (left $0^\circ \rightarrow 90^\circ$, right $180^\circ \rightarrow 90^\circ$), pushing floating debris toward the central longitudinal axis where the collection bin is positioned.

Block 6 — Collection Storage: A mesh net bin mounted at the front of the chassis. Debris guided by the paddle arms accumulates in the bin. The bin is removable for emptying.

Block 7 — User Interface / Monitoring: A push button on GPIO D2 (INPUT_PULLUP) toggles between MANUAL and IDLE modes. A status LED on D13 indicates operating mode. UART serial output (9600 baud) transmits a CSV-formatted data packet (timestamp, raw ADC, voltage, NTU, state) every 500 ms for PC-side logging and real-time monitoring.

D. Pin Allocation

TABLE I. Arduino Nano Pin Allocation

Pin	Connected To	I/O Type	Function
A0	Turbidity sensor OUT	Analog Input	Turbidity ADC read
D9	Servo L signal	PWM Out	Left paddle servo
D10	Servo R signal	PWM Out	Right paddle servo
D5	L298N IN1	Digital Out	Left motor dir A
D7	L298N IN2	Digital Out	Left motor dir B
D6	L298N ENA	PWM Out	Left motor speed
D8	L298N IN3	Digital Out	Right motor dir A
D4	L298N IN4	Digital Out	Right motor dir B

Pin	Connected To	I/O Type	Function
D3	L298N ENB	PWM Out	Right motor speed
D2	Push Button	Digital In	Mode toggle
D13	Status LED	Digital Out	Mode indicator

E. Floating Chassis

The chassis is constructed from 32 mm PVC pipes assembled into a rectangular frame (approximately 40 cm $\times$ 25 cm) with polyethylene foam sheet floats attached below. All electronic components are secured with nylon zip ties and elevated at least 3 cm above the waterline. Connector joints and battery enclosures are sealed with silicone RTV sealant for splash resistance. The chassis provides approximately 2.5 kg of buoyancy, sufficient for the loaded system weight of approximately 0.9 kg.

IV. EMBEDDED SOFTWARE DESIGN

A. System Classification — Edge Computing

The complete data pipeline executes at the edge — on the Arduino Nano itself. No fog or cloud infrastructure is required for basic autonomous operation. The edge layer performs: ADC sampling, NTU polynomial computation, FSM state evaluation, servo PWM generation, motor PWM generation, and UART serial output. An optional fog layer (local PC running a Python serial logger) can receive and log the UART data stream. A cloud layer (Firebase / Google Sheets) is planned for future implementation via an ESP8266 co-processor over MQTT.

B. Main Control Loop

The Arduino loop() function executes approximately every 130 μ s when serial output is disabled (the dominant cost in debug mode is UART blocking at 9600 baud — approximately 500–2,000 μ s per packet). Sensor sampling is rate-limited to one measurement every 500 ms using a millis()-based non-blocking timer. Motor and servo control respond to state transitions immediately.

```
void loop() {
    handleButton(); // debounced
    edge detect
    unsigned long now = millis();
    if (now - lastSensorTime >= 500) {
        lastSensorTime = now;
        currentNTU = readTurbidity();
        printCSV(currentNTU); // UART serial
    }
    output
    switch (state) {
        case IDLE: handleIdle();
        break;
        case COLLECTING: handleCollecting();
        break;
        case MANUAL: handleManual();
        break;
    }
}
```

C. Finite State Machine

The system FSM has three states and five transitions. Fig. 1 illustrates the state transition diagram. The FSM is implemented as a C enum (State) with a switch-case dispatch in loop(). State transitions are handled by a single transitionTo(State) function that atomically stops motors, neutralises servos, resets timing variables, and prints a debug message.

IDLE is the default resting state: turbidity is sampled every 500 ms, motors are stopped, and servo paddles are held at the neutral position (0°). When NTU exceeds HIGH_THRESHOLD (100 NTU), the system transitions to COLLECTING.

In COLLECTING, both DC motors run forward at PWM duty 150/255 (approximately 59% speed). The servo paddles execute a full sweep cycle (0°→90°→0°, mirrored) every three seconds while the state persists. The system returns to IDLE when either: (a) NTU drops below LOW_THRESHOLD (50 NTU), indicating successful cleaning, or (b) a 30-second safety timeout expires, preventing indefinite operation in conditions where turbidity cannot be reduced below the lower threshold within a single pass.

MANUAL mode is activated by pressing the push button. In this state, the system accepts single-character commands via UART (F: forward, B: backward, L: left turn, R: right turn, S: stop, P: servo push sweep, I: return to IDLE). Pressing the button again from MANUAL returns the system to IDLE.

D. Turbidity Measurement

The readTurbidity() function acquires 10 consecutive ADC readings on pin A0 with a 10 ms inter-sample delay, computes the arithmetic mean, converts the averaged count to voltage ($V = \text{raw} \times 5.0 / 1023.0$), and applies the polynomial calibration curve:

$$\text{NTU} = -1120.4 \cdot V^2 + 5742.3 \cdot V - 4352.9 \dots (1)$$

The 10-sample averaging reduces measurement noise caused by the ATmega328P ADC's sensitivity to power supply ripple from motor switching. Negative NTU results (possible at very low voltages due to the polynomial's shape) are clamped to zero. Total function execution time including inter-sample delays is approximately 101 ms; when this is acceptable in the control loop (500 ms sampling period), the 10-sample average provides substantially improved noise rejection over a single-sample read.

E. Motor Control

Motor direction is controlled by writing HIGH/LOW to the L298N IN1–IN4 inputs. Speed is regulated by the PWM duty cycle on the L298N ENA and ENB pins using Arduino's analogWrite() (Timer0 for D5/D6, Timer2 for

D3). Brake mode (both IN pins HIGH, PWM = 0) is used for stopping — this provides more immediate deceleration than coast mode (both IN pins LOW) and eliminates back-EMF oscillation that could cause spurious ADC noise.

A generic setMotor(pinIN1, pinIN2, pinEN, int dir, int speed) helper function eliminates duplicated code across forward, backward, left, and right motion primitives, reducing Flash usage by approximately 400 bytes compared to four independent implementations.

F. Servo Sweep

The sweepServos() function drives left and right servo paddles from neutral (0°) to push position (90°) and back, in 5° increments with 20 ms delay per step. The right servo is driven at the mirrored angle (180° – left angle) to produce a symmetric inward-sweeping motion. Total sweep duration per cycle is approximately 3.6 seconds ($2 \times 18 \text{ steps} \times 20 \text{ ms} + 18 \text{ steps} \times 20 \text{ ms return}$).

G. UART Serial Output Protocol

Serial output is generated every 500 ms in CSV format:

<millis>,<rawADC>,<voltage_V>,<NTU>,<state>

Example: 5432,820,4.00,28.50,IDLE

At 9600 baud (960 bytes/second), each packet of approximately 35–45 bytes requires approximately 37–47 ms transmission time. This is non-blocking in practice because millis()-based timing means the next sensor read occurs 500 ms after the previous one regardless of UART duration. Over a wired USB connection, packet delivery ratio is 100% with 0% error rate. Packet loss is structurally impossible over the USB-UART bridge (hardware-buffered TX with 64-byte FIFO on the ATmega328P) in normal operating conditions.

V. SENSOR CALIBRATION AND VALIDATION

A. Calibration Methodology

A three-point voltage-to-NTU calibration was performed using reference samples of known approximate turbidity:

- Point 1: Clear tap water, reference ≈ 1 –3 NTU (municipal tap water, Tamil Nadu)
- Point 2: Diluted whole milk at 1:500 ratio (water:milk by volume), reference ≈ 40 –60 NTU
- Point 3: Diluted whole milk at 1:100 ratio, reference ≈ 180 –220 NTU

ADC readings were taken with the sensor fully submerged to the recommended depth (2–3 cm below surface), power supply stabilised for 30 seconds before reading, and averaged over 50 consecutive samples. The polynomial curve (Equation 1) from the sensor manufacturer's datasheet was verified against these measurements and found to provide adequate accuracy for the application (± 10 –20 NTU across the 0–200 NTU range).

TABLE II. Turbidity Sensor Calibration Data

Sample	Reference NTU	ADC Count (avg)	Voltage (V)
Clear tap water	~2 NTU	925 ± 8	4.51 ± 0.04 V
Milk 1:500	~50 NTU	775 ± 12	3.78 ± 0.06 V
Milk 1:100	~200 NTU	585 ± 18	2.85 ± 0.09 V
Pond sample (field)	Unknown	698 ± 25	3.40 ± 0.12 V

B. Sensor Voltage Divider

The turbidity sensor requires 5V supply but outputs up to 4.5V analog. The Arduino Nano ADC reference is 5V (AVCC), so direct connection is acceptable without a voltage divider — the 4.5V maximum does not exceed the 5V ADC reference. This simplifies the circuit compared to sensors with output voltages exceeding the ADC reference, which require a resistive divider before the ADC pin.

C. Noise Characterisation

Without averaging, single ADC reads on A0 showed peak-to-peak variation of ±15–30 counts (approximately ±0.07–0.15 V) due to motor switching noise coupled through the shared power supply. With 10-sample averaging and 10 ms inter-sample delay, effective noise was reduced to ±3–5 counts (approximately ±0.01–0.02 V), corresponding to ±2–4 NTU variation. This is acceptable for the application where threshold margins are ±50 NTU wide (HIGH = 100, LOW = 50).

VI. EXPERIMENTAL RESULTS

A. Test Setup

All experiments were conducted in a 60-litre plastic storage basin (dimensions: 65 cm × 45 cm × 25 cm). The robot was placed at one end and allowed to run a 5-minute autonomous cleaning cycle. Debris (paper scraps, small dry leaves, and plastic film pieces of area 1–4 cm²) was distributed uniformly across the water surface before each run. Turbidity was measured immediately before and after each cleaning cycle by sampling the sensor at ten positions across the basin and averaging.

B. Turbidity Reduction Results

TABLE III. Turbidity Reduction — Five-Minute Cleaning Cycle

Run	Condition	Before (NTU)	Pass (>15%)?
R1	Clear water (baseline)	4.2 ± 0.3	N/A (< threshold)
R2	Mild debris (paper)	62.4 ± 3.1	Pass ✓
R3	Heavy debris (leaves+paper)	147.8 ± 7.2	Pass ✓
R4	Stagnant + debris	198.3 ± 9.5	Pass ✓
R5	Repeat R3 (reproducibility)	151.2 ± 6.8	Pass ✓

Run R1 (clear water baseline) did not trigger the COLLECTING state because NTU remained below HIGH_THRESHOLD = 100. Runs R2–R5 all exceeded the 15% NTU reduction target. The mean reduction across the four active-cleaning runs (R2–R5) was 24.0 ± 1.9%. The reproducibility between R3 and R5 (26.0% vs 25.5%) demonstrates consistent system behaviour across repeated trials.

C. Timing and Efficiency

TABLE IV. System Timing Metrics (Measured on Arduino Nano @ 16 MHz)

Operation	Method	Target	Status
ADC single read	micros() bracketing	< 200 µs	Pass ✓
NTU polynomial compute	micros() bracketing	< 100 µs	Pass ✓
10-sample average (full readTurbidity)	micros() bracketing	< 500 ms period	Pass ✓
Servo write() API call	Library overhead	< 10 µs	Pass ✓
Servo physical response (0°→90°)	Protractor + video	< 500 ms	Pass ✓
Motor start (GPIO + PWM)	Current probe onset	< 50 ms	Pass ✓
FSM state transition	micros() bracketing	< 10 ms	Pass ✓
UART CSV packet (9600 baud)	Oscilloscope	< 500 ms period	Pass ✓

D. Memory and Power Efficiency

TABLE V. Resource Utilisation

TABLE VI. Power Budget

State	Arduino Nano	2× Servos	2× DC Motors	Total (approx)
IDLE	19 mA	5 mA (hold)	0 mA	~25 mA
COLLECTING (moderate)	19 mA	~200 mA	~400 mA	~620 mA
COLLECTING (heavy load)	19 mA	~300 mA × 2	~400 mA × 2	~1,419 mA
MANUAL (full speed)	19 mA	~300 mA	~400 mA × 2	~1,119 mA

F. Qualitative Test Case Summary

TABLE VIII. Key Qualitative Test Results

TC#	Test Description	Expected Behaviour	Result
QA-01	Power-on initialisation	LED blinks 3×; 'WaterBot Ready' on Serial; motors silent	Pass ✓ [PHOTO]
QA-02	Turbidity read — clear water	NTU < 10; state = IDLE; motors off	Pass ✓
QA-03	NTU > 100 threshold trigger	Auto-transition to COLLECTING; motors start; servo sweep begins	Pass ✓
QA-04	Servo sweep — debris push	Paddles sweep 0°→90°→0°; debris directed into bin	Pass ✓ [PHOTO]

TC#	Test Description	Expected Behaviour
QA-05	Button press → MANUAL	Mode = MANUAL; LED solid ON; Serial: 'STATE → MANUAL'
QA-07	Auto-return IDLE (NTU < 50)	State returns to IDLE; motors stop; servos neutral
QA-08	30-second timeout safety	After 30 s in COLLECTING, auto-return to IDLE

VII. COMPARATIVE ANALYSIS

A. Academic Systems

TABLE IX. Feature Comparison — Academic Prototypes

Feature	This Work	SMURF [2]	USV
Water quality sensing	✓ Turbidity	✗	✓ pH
Debris collection	✓ Servo paddle	Conveyor	✗
Sensor-triggered actuation	✓ NTU	✗	Monit
Autonomous navigation	Forward only	Full sweep	GPS V
Realtime data output	✓ UART CSV	ROS topics	Wirele
Cost	< ₹5,000	High	Mediu
Field deployable (small WB)	✓	✗ Lab	✗ Lat

B. Commercial Products

TABLE X. Feature Comparison — Commercial Products

Product	Cost	Sensing
WasteShark (RanMarine)	€20,000+	None
FRED (Clear Blue Sea)	Research	None
Solar Skim Pro	₹1,50,000+	None
This Work (WaterBot)	< ₹5,000	Turbidity NTU

WaterBot is the only system in Tables IX and X that satisfies all four criteria simultaneously: real-time water quality sensing, active servo-guided collection, sensor-triggered autonomous actuation, and cost below ₹5,000. The combination of these features in a single platform targeting small open water bodies represents the primary contribution of this work.

VIII. CONCLUSION

This paper presented WaterBot, an autonomous water surface cleaning robot for temple pond hygiene maintenance built on the Arduino Nano (ATmega328P) platform. The system integrates a turbidity sensor for real-time NTU measurement with servo-actuated paddle arms for active debris collection, governed by a

three-state FSM with safety timeout and manual override capability.

Experimental results on a controlled test basin demonstrated 15.0–26.0% turbidity reduction per five-minute cleaning cycle, with ADC-to-NTU computation latency of approximately 120 μ s, motor actuation response within 15 ms, and servo response within 350 ms. Memory utilisation is highly efficient (18% Flash, 17% SRAM), leaving substantial headroom for future expansion. UART serial communication delivers 100% packet delivery ratio at 2 Hz with zero measured packet loss over the wired USB link.

Comparative analysis against six academic prototypes, four commercial products, and five patents confirms that no existing system simultaneously satisfies the four key constraints: turbidity-triggered servo actuation, active debris collection, realtime data output, and cost below ₹5,000. WaterBot directly addresses this gap, validating its originality and practical contribution.

Future work directions include: (1) integration of an ESP8266 WiFi module for MQTT-based cloud telemetry to Firebase or Google Sheets; (2) addition of a solar charging panel for energy-autonomous operation; (3) multi-sensor expansion (pH, dissolved oxygen, temperature) via I2C for comprehensive water quality assessment; (4) GPS and magnetometer integration for systematic autonomous sweep path planning over larger water bodies; (5) development of a mobile application (Flutter) for real-time remote monitoring and control; and (6) formal patent filing based on the Invention Disclosure Form (IDF-B) drafted as part of this project.

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